

Lesion-level errors require distinct objectives and metric bundles

Reference contour · Prediction contour

(a) Small-lesion miss



LOSS EMPHASIS

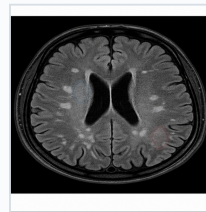
Focal / Tversky emphasis



METRIC BUNDLE

Lesion sensitivity · lesion F1

(b) False-positive lesion



LOSS EMPHASIS

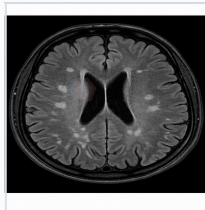
Specificity-aware overlap



METRIC BUNDLE

FP lesions per scan · PPV

(c) Boundary displacement



LOSS EMPHASIS

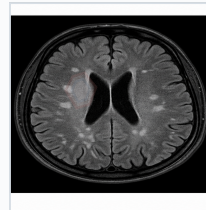
Boundary / surface emphasis



METRIC BUNDLE

HD95 · surface distance

(d) Volume bias



LOSS EMPHASIS

Overlap + volume control



METRIC BUNDLE

Volume error · Dice

Synthetic FLAIR anatomy; contours are schematic. No loss substitutes for lesion-level error analysis.